

Splunk SOC Lab — Windows Authentication Monitoring and Log Analysis

Overview

This project documents my hands-on learning experience building a beginner Security Operations Center (SOC) lab using Splunk Enterprise in a virtualized environment.

The objective of the lab was to gain practical exposure to:

- SIEM fundamentals
- Log ingestion and monitoring
- Splunk Search Processing Language (SPL)
- Windows authentication analysis
- Security event investigation
- Suspicious login detection workflows

The lab environment was built using Ubuntu, Windows, and Kali Linux virtual machines.

Lab Environment

Virtual Machines Used

Ubuntu VM

Used as the Splunk Enterprise server.

Windows VM

Used to generate Windows event logs and authentication activity.

Kali Linux VM

Used to simulate attacker activity and network logon attempts.

Network Configuration

All virtual machines were configured using:

- Adapter 1: NAT
- Adapter 2: Host-Only Adapter

This setup allowed:

- Internet connectivity
 - Internal VM communication
 - Controlled SOC lab simulations
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Technologies and Tools

Tool	Purpose
Splunk Enterprise	SIEM and log analysis
Splunk Universal Forwarder	Log forwarding
Sysmon	Advanced Windows telemetry
Ubuntu Linux	Splunk server hosting
Windows	Event generation
Kali Linux	Simulated attacker system
VirtualBox / VMware	Virtualization

Splunk Installation and Configuration

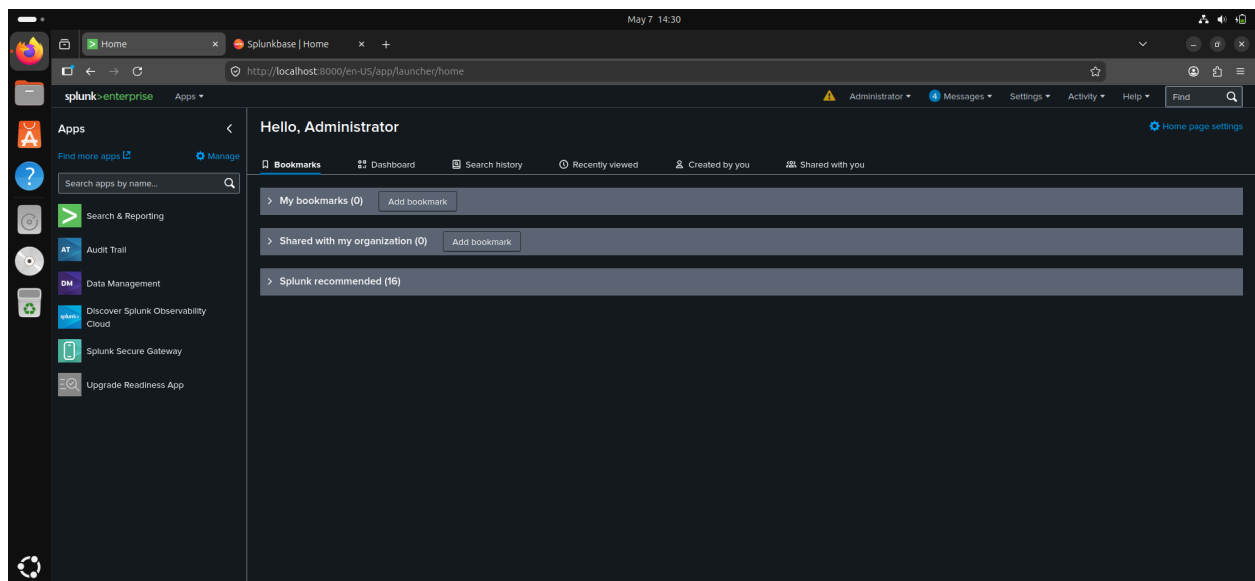
Splunk Enterprise was installed and configured on the Ubuntu virtual machine.

After installation, I:

- Started the Splunk service
- Accessed the Splunk web interface
- Configured data ingestion
- Practiced basic SPL searches
- Explored indexes and source types

The installation process helped me understand how SIEM platforms centralize and analyze logs.

Screenshot 1 — Splunk Home Dashboard



Log Ingestion and SPL Practice

To begin learning log analysis, I ingested logs into Splunk and practiced using SPL queries.

Some of the searches practiced included:

`index=*`

`index=* sourcetype=*`

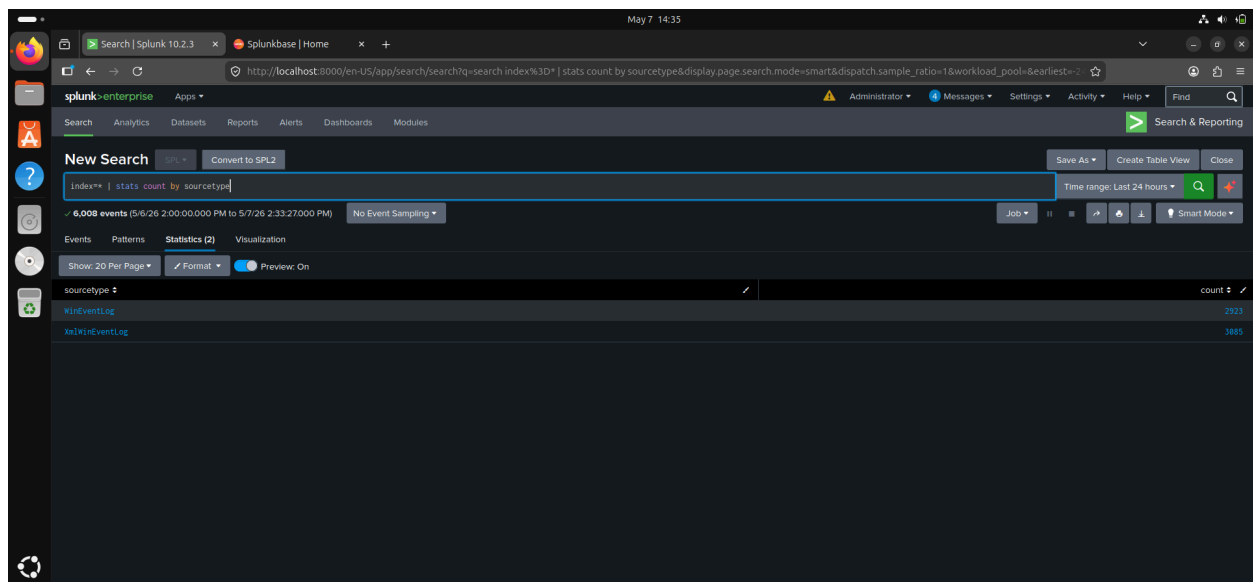
`index=* | stats count by sourcetype`

index=* | head 20

This phase helped me understand:

- Searching across indexes
 - Understanding source types
 - Viewing raw events
 - Filtering data
 - Event structure analysis
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Screenshot 2 — Basic SPL Search Results



Configuring Sysmon and Splunk Forwarder

To improve endpoint visibility and centralize Windows logs, I installed:

- Sysmon on the Windows virtual machine
- Splunk Universal Forwarder on the Windows virtual machine

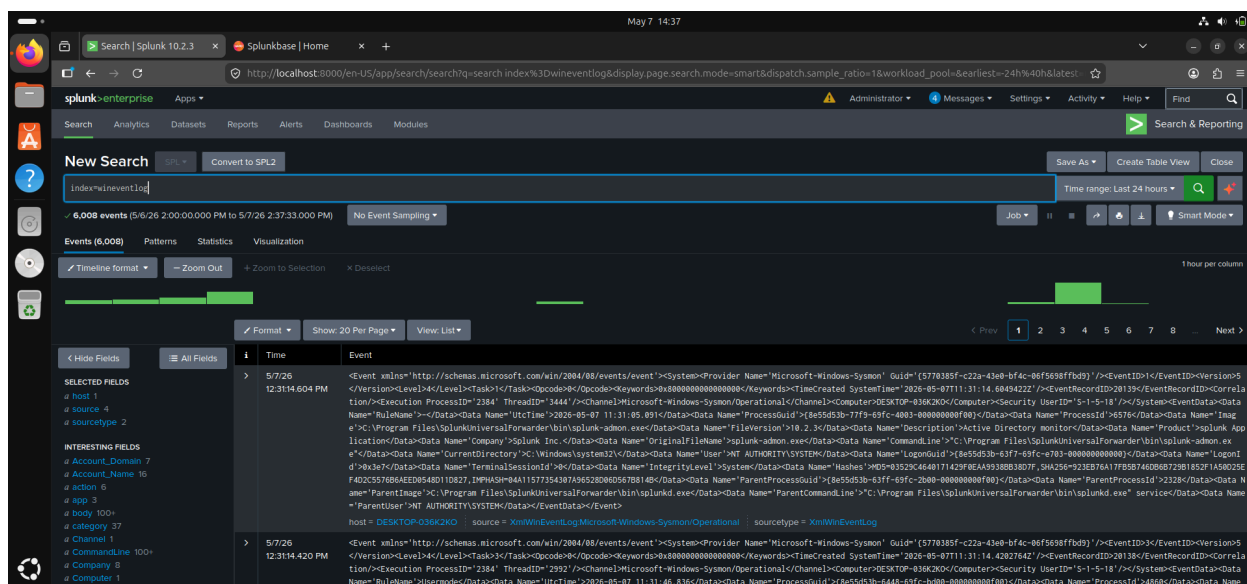
The Splunk Forwarder was configured to send Windows logs to the Splunk server hosted on Ubuntu.

This allowed the SIEM to ingest:

- Windows Security logs
- Authentication events
- Endpoint activity
- System telemetry generated by Sysmon

This setup introduced me to the concept of centralized logging and endpoint monitoring commonly used in real SOC environments.

Screenshot 3 — Windows Logs Successfully Forwarded to Splunk



Authentication Monitoring and Suspicious Activity Detection

One of the most important parts of this lab involved investigating authentication events.

Using the Kali Linux machine, I generated network authentication attempts against the Windows system.

The logs collected in Splunk showed:

- Multiple failed login attempts
- A successful authentication afterward
- Logon Type 3 network logons
- Source IP address information
- Workstation name identified as “kali”

This pattern is significant because multiple failed logons followed by a successful login may indicate:

- Brute-force attacks
 - Password spraying
 - Credential guessing
 - Unauthorized access attempts
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SPL Query Used for Investigation

```
index=* EventCode=4625 OR EventCode=4624
```

Investigation Findings

Event ID 4625 — Failed Logons

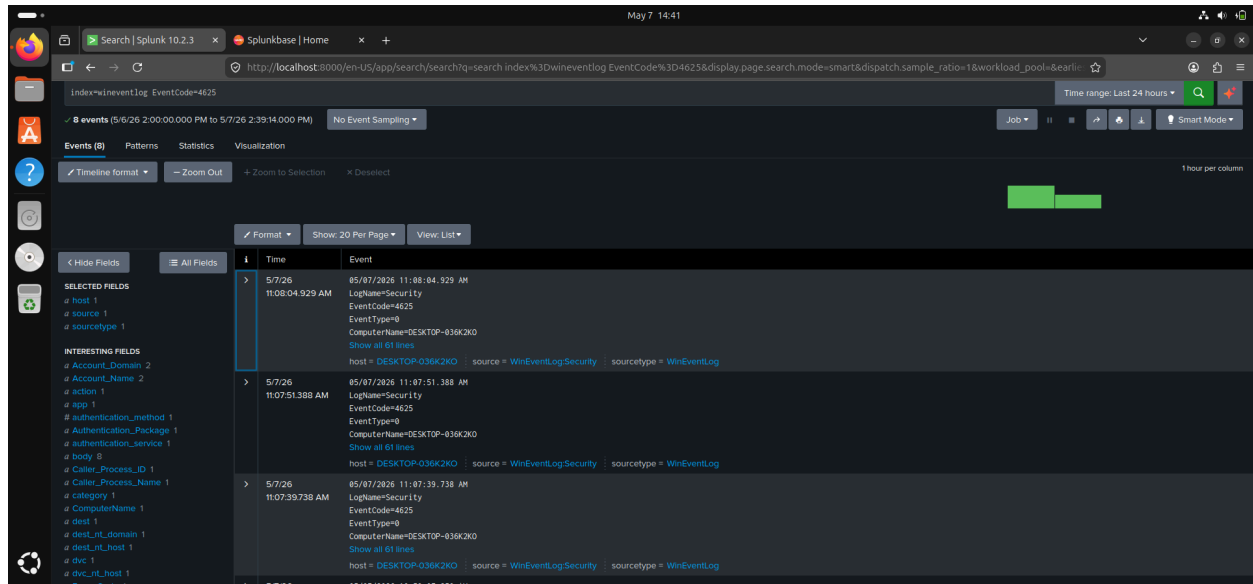
I observed several failed authentication attempts before a successful login occurred.

These events provided:

- Failed username attempts
- Timestamps
- Authentication details
- Source activity information

This demonstrated how failed authentication events can be used to identify suspicious behavior.

Screenshot 4 — Windows Event ID 4625 Failed Logons



Event ID 4624 — Successful Logon

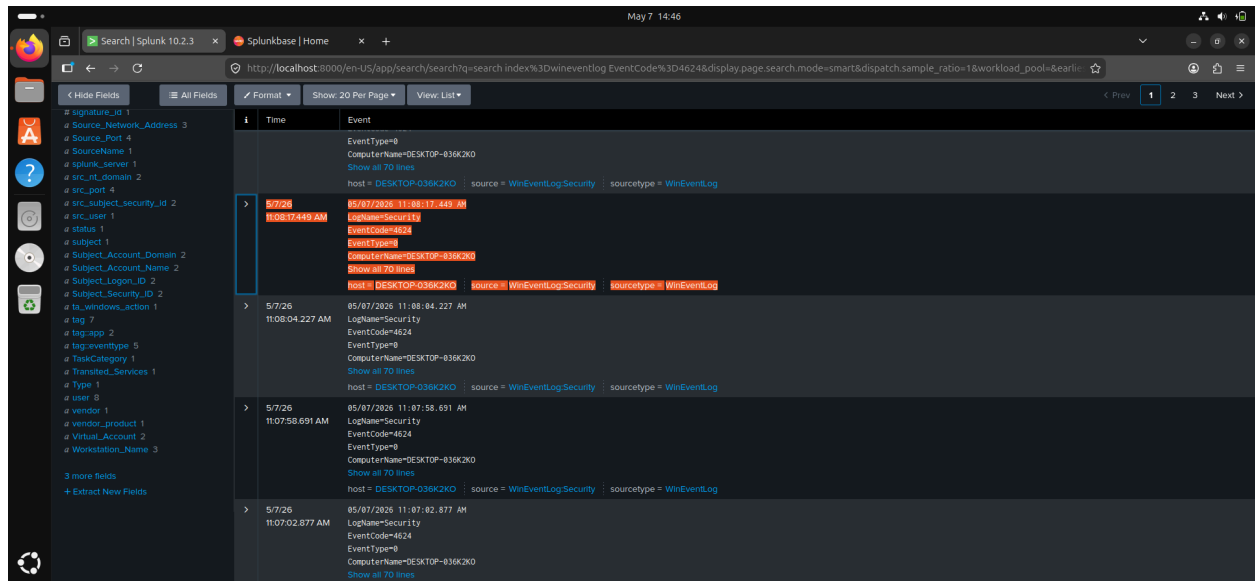
I also analyzed Event ID 4624, which represents successful authentication activity.

The event revealed:

- Logon Type 3 (network logon)
- Source IP address from the Kali machine
- Workstation name “kali”
- Successful authentication after failed attempts

This exercise helped me understand how SOC analysts correlate events during investigations.

Screenshot 5 — Windows Event ID 4624 Successful Network Logon



Skills and Knowledge Gained

Through this project, I gained practical experience in:

- Splunk installation and configuration
 - SIEM fundamentals
 - Windows log analysis
 - SPL query writing
 - Log forwarding
 - Authentication monitoring
 - Event correlation
 - Basic threat detection workflows
 - Investigating suspicious login behavior
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Challenges Encountered

Some challenges encountered during the project included:

- Configuring VM networking
- Ensuring all systems communicated properly
- Understanding Windows Event IDs

- Learning SPL syntax
- Troubleshooting log ingestion
- Configuring forwarders correctly

Resolving these issues improved both my troubleshooting and analytical skills.

Future Improvements

Future improvements planned for the lab include:

- Creating custom Splunk dashboards
 - Building correlation searches
 - Developing authentication alerts
 - Expanding Sysmon monitoring coverage
 - Simulating additional attack scenarios
 - Adding more endpoint log sources
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Conclusion

This project helped me transition from theoretical learning into practical SOC analysis.

By building a functional Splunk lab environment and investigating authentication logs, I developed foundational SIEM and security monitoring skills relevant to entry-level SOC analyst roles.

The project also strengthened my understanding of:

- Security event analysis
- Authentication investigations
- Windows logging
- SIEM workflows
- Threat detection concepts
- Security monitoring operations

This lab will continue evolving as I expand my cybersecurity and SOC analysis skills.

Author

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